O2 Number theory — notes

← Back to the module README

Number theory (getaltheorie) is about the integers (gehele getallen): which numbers divide which, what remainders are left over, and which equations have whole-number solutions. Olympiad number theory uses a small set of tools, introduced here one by one. Try each worked example yourself before reading its solution.

Notation: $a \mid b$ means "a divides b" (that is, $b = ak$ for some integer k). $\gcd(a, b)$ is the greatest common divisor and $\operatorname{lcm}(a, b)$ the least common multiple. $S(n)$ is the digit sum (cijfersom) of n.

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1. Divisibility (deelbaarheid)

Idea. Divisibility statements can be added, subtracted and multiplied, just like equations. That is the engine behind most of this module.

Key facts.

1. If $d \mid a$ and $d \mid b$, then $d \mid ax + by$ for all integers x, y. *Proof.* $a = dk$, $b = dl$, so $ax + by = d(kx + ly)$.
2. If $a \mid b$ and $b \mid c$ then $a \mid c$.
3. If $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$. (This "size" fact is often how a proof finishes.)

Divisibility rules (deelbaarheidsregels). Write $n = \overline{a_k \ldots a_1 a_0}$ in decimal, so $n = a_0 + 10a_1 + 100a_2 + \ldots$

- By 2, 5, 10: look at the last digit (because $10 \mid n - a_0$).
- By 4, 25: the last two digits; by 8, 125: the last three digits (because $100 = 4 \cdot 25$ and $1000 = 8 \cdot 125$).
- By 3 and 9: the digit sum. *Proof.* $10 = 9 + 1$, so $10^j - 1 = 99 \ldots 9$ is a multiple of 9. Hence $n - S(n) = a_1(10 - 1) + a_2(100 - 1) + \ldots$ is a multiple of 9. So n and $S(n)$ leave the same remainder on division by 9 (and by 3).
- By 11: the alternating digit sum $a_0 - a_1 + a_2 - \ldots$ *Proof.* $10 = 11 - 1$, so $10^j - (-1)^j$ is a multiple of 11 (check: $10 + 1 = 11$, $100 - 1 = 99$, $1000 + 1 = 1001 = 7 \cdot 11 \cdot 13$).
- By a composite number like 36: split it into coprime factors, $36 = 4 \cdot 9$, and test each.

Example 1.1 (classic). Take any three-digit number $\overline{abc}$ and write it twice: $\overline{abcabc}$. Show the result is always divisible by 7, 11 and 13.

Solution. $\overline{abcabc} = 1000 \cdot \overline{abc} + \overline{abc} = 1001 \cdot \overline{abc}$, and $1001 = 7 \cdot 11 \cdot 13$.

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2. Primes and prime factorisation (priemgetallen en priemfactorontbinding)

Idea. A prime (priemgetal) is an integer $p > 1$ whose only positive divisors are 1 and p. Primes are the atoms: every integer is built from them in exactly one way.

Fundamental theorem of arithmetic (hoofdstelling van de rekenkunde). Every integer $n > 1$ can be written as a product of primes, and this product is unique apart from the order of the factors.

Existence. If n is prime we are done. Otherwise $n = ab$ with $1 < a, b < n$, and we factor a and b (strong induction). *Uniqueness* needs Euclid's lemma (section 5): if a prime p divides ab, then $p \mid a$ or $p \mid b$. Given two factorisations of n, a prime in the first one divides the product of the second, so by Euclid's lemma it equals one of those primes; cancel it and repeat.

Theorem (Euclid). There are infinitely many primes. *Proof.* Suppose $p_1, \dots, p_k$ were all the primes. Look at $N = p_1 p_2 \cdots p_k + 1$. It has some prime factor q. But N leaves remainder 1 when divided by each p_i, so q is not on the list. Contradiction.

Useful facts.

- To test whether n is prime, it is enough to try primes $p \le \sqrt{n}$ (if $n = ab$ with $a \le b$, then $a \le \sqrt{n}$).
- Every prime $p > 3$ is of the form $6k \pm 1$, because $6k, 6k \pm 2, 6k + 3$ are divisible by 2 or 3.
- "Look modulo 3" is the standard first move when a problem asks for primes p such that several expressions in p are prime.

Example 2.1. Find all primes p such that $p + 10$ and $p + 14$ are also prime.

Solution. Look at remainders on division by 3. If p leaves remainder 1, then $p + 14$ leaves remainder 0, so $3 \mid p + 14$, and $p + 14 > 3$, not prime. If p leaves remainder 2, then $3 \mid p + 10$, not prime. So $3 \mid p$, which means $p = 3$, and indeed 13 and 17 are prime. **Answer:** $p = 3$**.**

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3. Number of divisors (aantal delers)

Idea. Once you know the prime factorisation, every divisor is a choice of exponents.

Formula. If $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, then the divisors of n are exactly the numbers $p_1^{f_1} \cdots p_k^{f_k}$ with $0 \le f_i \le e_i$. There are $e_i + 1$ choices for each f_i, so

$$\tau(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1).$$

(The letter τ is "tau"; some books write $d(n)$.) For example $72 = 2^3 \cdot 3^2$ has $4 \cdot 3 = 12$ divisors.

Fact. $\tau(n)$ is odd exactly when n is a perfect square. *Proof 1:* all $e_i + 1$ odd means all e_i even. *Proof 2 (pairing):* divisors come in pairs $d \leftrightarrow n/d$; a divisor is paired with itself only if $d^2 = n$.

Example 3.1 (locker problem, classic). A corridor has 100 closed lockers. Student k (for $k = 1, \dots, 100$) toggles every k-th locker. Which lockers are open at the end?

Solution. Locker n is toggled once for each divisor of n. It ends open when $\tau(n)$ is odd, i.e. when n is a square: lockers $1, 4, 9, \dots, 100$ — ten lockers.

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4. gcd, lcm and Euclid's algorithm (ggd, kgv en het algoritme van Euclides)

Idea. The greatest common divisor (grootste gemene deler, ggd) and least common multiple (kleinste gemene veelvoud, kgv) can be read off from prime factorisations: take the minimum, respectively the maximum, of each exponent. Since $\min(e, f) + \max(e, f) = e + f$,

$$\gcd(a, b) \cdot \operatorname{lcm}(a, b) = ab.$$

For large numbers, factorising is slow. Euclid's algorithm is fast and needs no factorisation.

Key lemma. $\gcd(a, b) = \gcd(a - qb, b)$ for every integer q. *Proof.* By fact 1 of section 1, any common divisor of a and b divides $a - qb$; and any common divisor of $a - qb$ and b divides $(a - qb) + qb = a$. So the two pairs have the same common divisors.

Euclid's algorithm. Divide with remainder repeatedly; the last non-zero remainder is the gcd.

Example 4.1. Compute $\gcd(1071, 462)$.

$$1071 = 2 \cdot 462 + 147$$
$$462 = 3 \cdot 147 + 21$$
$$147 = 7 \cdot 21 + 0$$

So $\gcd(1071, 462) = 21$.

Example 4.2 (classic, IMO 1959). Prove that the fraction $\dfrac{21n+4}{14n+3}$ cannot be simplified for any positive integer n.

Solution. Use the key lemma with algebraic expressions:

$$\gcd(21n+4,\ 14n+3) = \gcd(7n+1,\ 14n+3) = \gcd(7n+1,\ 1) = 1.$$

(First subtract the second from the first; then subtract $2 \cdot (7n+1)$ from $14n+3$.)

Example 4.3. Two positive integers have gcd 6 and lcm 72. What can they be?

Solution. Write $a = 6x$, $b = 6y$ with $\gcd(x, y) = 1$. Then $\operatorname{lcm}(a, b) = 6xy = 72$, so $xy = 12 = 2^2 \cdot 3$. Because x, y are coprime, each prime power goes wholly to one of them: $\{x, y\} = \{1, 12\}$ or $\{3, 4\}$. **Answer:** $\{6, 72\}$ **or** $\{18, 24\}$**.** This "write $a = gx$, $b = gy$ with x, y coprime" trick is worth memorising.

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5. Bézout's identity and linear Diophantine equations (stelling van Bézout)

Theorem (Bézout). For integers a, b, not both 0, there are integers x, y with $ax + by = \gcd(a, b)$. More precisely: the numbers of the form $ax + by$ are exactly the multiples of $\gcd(a, b)$.

Proof. Let d be the smallest positive number of the form $ax + by$. Divide a by d: $a = qd + r$ with $0 \le r < d$. Then $r = a - q(ax + by)$ is again of the form $ax' + by'$, and $r < d$, so by minimality $r = 0$, i.e. $d \mid a$. Likewise $d \mid b$. On the other hand every common divisor of a, b divides $ax + by = d$. So $d = \gcd(a, b)$. Every multiple of d is then also of the form $ax + by$, and every number $ax + by$ is a multiple of $\gcd(a, b)$.

Finding x, y: run Euclid backwards (extended Euclid). For $\gcd(1071, 462) = 21$:

$$21 = 462 - 3 \cdot 147 = 462 - 3(1071 - 2 \cdot 462) = 7 \cdot 462 - 3 \cdot 1071.$$

Euclid's lemma. If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally: if $n \mid ab$ and $\gcd(n, a) = 1$ then $n \mid b$. *Proof.* Write $1 = nx + ay$. Multiply by b: $b = nbx + (ab)y$. Both terms are multiples of n, so $n \mid b$. (For the prime case: if $p \nmid a$ then $\gcd(p, a) = 1$.)

Linear Diophantine equations (lineaire diophantische vergelijkingen). The equation $ax + by = c$ has integer solutions if and only if $\gcd(a, b) \mid c$. If (x_0, y_0) is one solution and $g = \gcd(a, b)$, then all solutions are

$$x = x_0 + \frac{b}{g}t, \qquad y = y_0 - \frac{a}{g}t, \qquad t \in \mathbb{Z}.$$

Why all? If $ax + by = c = ax_0 + by_0$, then $a(x - x_0) = b(y_0 - y)$. Divide by g and use Euclid's lemma (a/g and b/g are coprime): $\frac{b}{g} \mid x - x_0$.

Example 5.1. Solve $17x + 29y = 1$ in integers.

Solution. Euclid: $29 = 1 \cdot 17 + 12$, $17 = 1 \cdot 12 + 5$, $12 = 2 \cdot 5 + 2$, $5 = 2 \cdot 2 + 1$. Backwards: $1 = 5 - 2 \cdot 2 = 5 - 2(12 - 2 \cdot 5) = 5 \cdot 5 - 2 \cdot 12 = 5(17 - 12) - 2 \cdot 12 = 5 \cdot 17 - 7 \cdot 12 = 5 \cdot 17 - 7(29 - 17) = 12 \cdot 17 - 7 \cdot 29$. So $(x_0, y_0) = (12, -7)$ and **all solutions are** $x = 12 + 29t$**,** $y = -7 - 17t$**.**

Small coefficients in a contest? Often it is quicker to reduce modulo the smaller coefficient: in $5x + 8y = 100$, look mod 5: $8y \equiv 0$, so $5 \mid y$; try $y = 0, 5, 10$ to get the non-negative solutions $(20, 0), (12, 5), (4, 10)$.

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6. Modular arithmetic (modulorekenen, restklassen)

Idea. Often we only care about remainders. Write $a \equiv b \pmod{n}$ ("a is congruent to b modulo n") when $n \mid a - b$, i.e. a and b leave the same remainder on division by n.

Rules. If $a \equiv b$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd, \qquad a^k \equiv b^k \pmod{n}.$$

Proof of the product rule. $ac - bd = a(c - d) + d(a - b)$, and both terms are multiples of n.

Warning: division. $2 \cdot 3 \equiv 2 \cdot 8 \pmod{10}$ but $3 \not\equiv 8 \pmod{10}$. You may cancel a factor c only if $\gcd(c, n) = 1$ (by Euclid's lemma). If $\gcd(a, n) = 1$, Bézout gives $ax + ny = 1$, so $ax \equiv 1 \pmod{n}$: x is the inverse of a modulo n.

The power trick. To compute a^N mod n, find a small power with $a^k \equiv \pm 1$ and use $a^{qk+r} = (a^k)^q \cdot a^r$.

Example 6.1. Find the remainder of 3^{100} on division by 7.

Solution. Powers of 3 mod 7: $3, 2, 6, 4, 5, 1$. So $3^6 \equiv 1$, and $3^{100} = (3^6)^{16} \cdot 3^4 \equiv 3^4 = 81 \equiv 4$. **Answer: 4.**

Example 6.2. Find the last two digits of 7^{100}.

Solution. "Last two digits" means "modulo 100". $7^2 = 49$, $7^4 = 2401 \equiv 1 \pmod{100}$. So $7^{100} = (7^4)^{25} \equiv 1$. **Answer: 01.**

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7. Squares and cubes modulo small numbers (kwadraten en derde machten modulo kleine getallen)

Idea. Squares can only leave certain remainders. If an equation forces a forbidden remainder, it has no solutions. Checking every residue class takes a minute; memorise this table:

modulus	possible remainders of x^2
3	0, 1
4	0, 1
5	0, 1, 4
8	0, 1, 4 (odd squares are $\equiv 1$)
9	0, 1, 4, 7
10	0, 1, 4, 5, 6, 9 (last digit of a square)

Cubes modulo 9 are only $0, 1, 8$ (that is $0, \pm 1$). Cubes modulo 7 are only $0, 1, 6$.

Proof that odd squares are $\equiv 1 \pmod 8$. $(2k+1)^2 = 4k(k+1) + 1$, and $k(k+1)$ is even.

How to pick the modulus. Squares: try 3, 4, 8, 5. Cubes: try 9 or 7. Powers of 2 or 3 in the equation: try a modulus where those powers repeat quickly (3, 4, 7, 8, 9).

Example 7.1. Show that $x^2 + y^2 = 2027$ has no integer solutions.

Solution. $x^2 + y^2 \pmod 4$ is one of $0+0, 0+1, 1+1$, so $0, 1$ or 2. But $2027 = 4 \cdot 506 + 3 \equiv 3$. Impossible.

Example 7.2. Show that $x^3 + y^3 + z^3 = 2030$ has no integer solutions.

Solution. Each cube is $\equiv 0, 1$ or $-1 \pmod 9$, so a sum of three cubes is between -3 and 3 modulo 9: remainders $0, 1, 2, 3, 6, 7, 8$. But $2030 \equiv 2 + 0 + 3 + 0 = 5 \pmod 9$.

Example 7.3. Can a number whose digit sum is 2027 be a perfect square?

Solution. No: such a number is $\equiv 2027 \equiv 2 \pmod 9$ (section 1), and squares are $0, 1, 4, 7 \pmod 9$.

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8. Fermat's little theorem (kleine stelling van Fermat)

Theorem. If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod p$. Equivalently, $a^p \equiv a \pmod p$ for every integer a.

Proof. Look at the $p-1$ numbers $a, 2a, 3a, \ldots, (p-1)a$ modulo p.

- None of them is $\equiv 0$: $p \mid ka$ with $p \nmid a$ would need $p \mid k$ (Euclid's lemma), but $1 \le k \le p-1$.
- No two are congruent: $ia \equiv ja$ means $p \mid (i-j)a$, so $p \mid i-j$, so $i = j$.

So their remainders are exactly $1, 2, \ldots, p-1$ in some order. Multiply all of them:

$$a \cdot 2a \cdots (p-1)a \equiv 1 \cdot 2 \cdots (p-1) \pmod p, \quad \text{i.e.} \quad (p-1)!\, a^{p-1} \equiv (p-1)! \pmod p.$$

Since $(p-1)!$ is coprime to p, we may cancel it: $a^{p-1} \equiv 1$. ■

Example 8.1. Find the remainder of 2^{2027} on division by 11.

Solution. Fermat: $2^{10} \equiv 1 \pmod{11}$. $2027 = 202 \cdot 10 + 7$, so $2^{2027} \equiv 2^7 = 128 = 11 \cdot 11 + 7$. **Answer: 7.**

Example 8.2. Prove that $n^7 - n$ is divisible by 42 for every integer n.

Solution. $42 = 2 \cdot 3 \cdot 7$. By Fermat ($a^p \equiv a$) with $p = 7$: $7 \mid n^7 - n$. For 3: $n^3 \equiv n$, so $n^7 = (n^3)^2 \cdot n \equiv n^2 \cdot n = n^3 \equiv n \pmod 3$. For 2: n^7 and n have the same parity. Since 2, 3, 7 are distinct primes, their product divides $n^7 - n$.

A taste of order (orde). If $\gcd(a, n) = 1$, the order of a modulo n is the smallest $d \ge 1$ with $a^d \equiv 1$. Key fact: $a^m \equiv 1 \pmod n$ if and only if $d \mid m$. (*Proof:* write $m = qd + r$ with $0 \le r < d$; then $a^m \equiv a^r$, and $a^r \equiv 1$ with $r < d$ forces $r = 0$.) Combined with Fermat: the order of a modulo a prime p divides $p - 1$.

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9. Digits and number bases (cijfers en talstelsels)

Idea. In base b (grondtal b), the string $\overline{a_k \ldots a_1 a_0}_b$ with digits $0 \le a_i \le b-1$ means $a_k b^k + \cdots + a_1 b + a_0$. Every positive integer has exactly one such representation.

Converting. Divide repeatedly by b and read the remainders from bottom to top. For 2026 in base 3: $2026 = 3 \cdot 675 + 1$, $675 = 3 \cdot 225 + 0$, $225 = 3 \cdot 75 + 0$, $75 = 3 \cdot 25 + 0$, $25 = 3 \cdot 8 + 1$, $8 = 3 \cdot 2 + 2$, $2 = 3 \cdot 0 + 2$. So $2026 = 2210001_3$. In base 2, $2026 = 11111101010_2$.

Facts.

- The last digit in base b is n mod b. The last k digits encode n mod b^k.
- n has exactly k digits in base b if and only if $b^{k-1} \le n < b^k$.
- In base b, the number and its digit sum are congruent modulo $b-1$ (same proof as for 9 in base 10).
- $S(n) \equiv n \pmod 9$, and $S(n) \le 9 \cdot (\text{number of digits})$. These two facts together solve most digit-sum problems: the congruence narrows down the residue class and the bound narrows down the size.

Example 9.1. Find all positive integers n with $n + S(n) = 100$.

Solution. $n < 100$ so $S(n) \le 18$ and $n \ge 82$. Mod 9: $n + S(n) \equiv 2n \equiv 100 \equiv 1$, so $n \equiv 5 \pmod 9$ (because $2 \cdot 5 = 10 \equiv 1$). Candidates in $[82, 99]$: $86, 95$. Check: $86 + 14 = 100$ works; $95 + 14 = 109$ does not. **Answer:** $n = 86$**.**

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10. Diophantine equations (diophantische vergelijkingen)

A Diophantine equation asks for integer solutions. Four standard weapons:

1. **Factoring.** Turn the equation into (product) = (fixed number), then list divisor pairs.
2. **Simon's favourite factoring trick (SFFT).** $xy+ax+by = c$ becomes $(x+b)(y+a) = c+ab$ by adding ab to both sides.
3. **Modular obstruction.** Show the two sides can never be congruent modulo some n (section 7).
4. **Bounding.** Show that large values make one side too big, then check the small cases.

Example 10.1. Find all positive integers x, y with $xy = x + y + 5$.

Solution. $xy - x - y + 1 = 6$, i.e. $(x-1)(y-1) = 6$. Both factors are ≥ 0 and their product is positive, so they are a positive divisor pair of 6: $(1,6), (2,3), (3,2), (6,1)$. **Answer:** $(x, y) = (2,7), (3,4), (4,3), (7,2)$**.**

Example 10.2. Find all non-negative integers x and positive integers y with $2^x + 7 = y^2$.

Solution. If $x \ge 2$, then $2^x + 7 \equiv 3 \pmod 4$, but squares are 0 or $1 \pmod 4$. So $x \in \{0, 1\}$. $x = 0$ gives 8, not a square; $x = 1$ gives $9 = 3^2$. **Answer:** $(x, y) = (1, 3)$**.**

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11. Infinite descent (oneindige afdaling)

Idea (Fermat). Suppose a solution in positive integers exists. Build from it a strictly *smaller* solution. Then you could keep going forever, giving an infinite decreasing sequence of positive integers — impossible. Formally: take a solution with the smallest possible value of some positive quantity, and produce an even smaller one. This is the well-ordering principle you met in O1 (extremal principle).

Example 11.1. Prove that $\sqrt{2}$ is irrational.

Solution. Suppose $\sqrt{2} = a/b$ with positive integers a, b, and choose the solution of $a^2 = 2b^2$ with b as small as possible. Then a^2 is even, so a is even: $a = 2c$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$: now (b, c) is a solution with $c < b$. Contradiction with minimality.

The same pattern — "a modulus forces every variable to be divisible by p, divide everything by p" — appears in problem 23.

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12. Legendre's formula and trailing zeros (formule van Legendre, nullen aan het eind)

Idea. How many times does a prime p divide $n! = 1 \cdot 2 \cdots n$? Count, for each k, how many of $1, \ldots, n$ are multiples of p^k.

Legendre's formula. The exponent of p in $n!$ is

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

where $\lfloor x \rfloor$ is x rounded down. *Proof.* A number $m \leq n$ with $p^j \mid m$ but $p^{j+1} \nmid m$ contributes j factors p. It is counted exactly j times on the right: once in each of the first j terms (there are $\lfloor n/p^k \rfloor$ multiples of p^k up to n).

Trailing zeros (nullen aan het eind). Each final zero of $n!$ needs a factor $10 = 2 \cdot 5$. There are always more 2s than 5s, so the number of trailing zeros of $n!$ is $v_5(n!)$.

Example 12.1. How many zeros does $100!$ end in, and what is the largest power of 3 dividing $100!$?

Solution. $v_5(100!) = 20 + 4 = 24$ zeros. $v_3(100!) = 33 + 11 + 3 + 1 = 48$, so 3^{48}.

Toolbox summary

Situation	First thing to try
Divisibility of a number written with digits	Digit rules: 9/3 (digit sum), 11 (alternating sum), 4/8 (last digits)
"Find all primes p such that ..."	Look modulo 3 (or 2, 5); the small prime is usually the only answer
Counting divisors / smallest number with k divisors	$\tau(n) = \prod(e_i + 1)$; large exponents on small primes
gcd of two expressions in n	Euclid: subtract multiples until a constant remains
Given gcd and lcm	Write $a = gx$, $b = gy$ with $\gcd(x, y) = 1$
$ax + by = c$	Solvable iff $\gcd(a, b)$ divides c; one solution plus $(b/g, -a/g)t$; or reduce mod the small coefficient
Remainder of a huge power	Find $a^k \equiv \pm 1$; Fermat gives $a^{p-1} \equiv 1 \pmod{p}$
"Show m divides $f(n)$ for all n"	Factor m into prime powers; check each residue class or use Fermat

Situation	First thing to try
Equation with squares or cubes, "no solutions"	Squares mod 3, 4, 5, 8; cubes mod 7, 9
$xy + ax + by = c$	SFFT: $(x + b)(y + a) = c + ab$
$x^2 - y^2 = c$ or powers of 2	Factor; the factors have the same parity / must be powers of 2
Digit sums	$S(n) \equiv n \pmod 9$ plus the bound $S(n) \leq 9\cdot$ (number of digits)
Numbers in base b	Expand as a polynomial in b; last digit $= n$ mod b
Only the zero solution seems possible	Infinite descent: divide all variables by p
Powers of a prime in $n!$, trailing zeros	Legendre: $\sum \lfloor n/p^k \rfloor$
Stuck	Small cases, find the pattern, prove it (O1)